Section 20: OSPF

Layer: L3 | Reference: Odom Vol1 Ch21-24: OSPF Concepts, Configuration, Neighbors | *Original study notes — CCNA 200-301*

OSPF (Open Shortest Path First) is a link-state Interior Gateway Protocol. Every router builds an identical Link-State Database (LSDB) describing the full topology, then independently runs Dijkstra's SPF algorithm to calculate the best path to every destination. It is the most heavily tested dynamic routing protocol on the CCNA exam.

Key Concepts

- Neighbor states progress: Down -> Init -> 2-Way -> ExStart -> Exchange -> Loading -> Full
- Administrative Distance = 110
- Cost (metric) = reference bandwidth / interface bandwidth. Default reference bandwidth = 100,000 Kbps (100 Mbps)
- Lower cost is preferred; OSPF sums the cost of every outgoing interface along the path
- DR (Designated Router) and BDR (Backup DR) are elected on multi-access segments (broadcast/NBMA) to reduce the number of adjacencies; point-to-point links skip DR/BDR election entirely
- DR/BDR election: highest OSPF priority wins (default 100, 0 = ineligible); tie broken by highest Router ID
- Router ID (RID) selection order: manually configured router-id > highest IP on an active loopback interface > highest IP on any active physical interface at process startup
- Hello and Dead timers must match between neighbors (default on Ethernet: Hello=10s, Dead=40s; default on point-to-point serial differs by media)
- OSPF uses multicast 224.0.0.5 (AllSPFRouters) and 224.0.0.6 (AllDRouters) for Hellos and updates
- Areas: Area 0 is the backbone; all other areas must connect to Area 0 directly or via a virtual link
- ABR (Area Border Router) sits between two areas; ASBR (Autonomous System Boundary Router) redistributes routes from outside OSPF
- LSA Types: Type 1 (Router LSA, within area), Type 2 (Network LSA, generated by DR), Type 3 (Summary LSA, generated by ABR between areas), Type 5 (External LSA, generated by ASBR)
- Passive interfaces still get advertised in OSPF but suppress Hello packets - used on LAN-facing interfaces with no other routers
- OSPFv2 routes IPv4; OSPFv3 routes IPv6 (separate protocol, similar mechanics)

Command Reference

Command	Purpose
`router ospf <process-id>`	Enter OSPF config mode; process ID is locally significant only, does not need to match between routers
`router-id <a.b.c.d>`	Manually set the Router ID (requires clear ip ospf process to take effect immediately)

Command	Purpose
network <ip> <wildcard-mask> area <id>	Enable OSPF on any interface whose IP matches; assigns it to the given area
passive-interface <intf>	Suppress OSPF Hellos on an interface while still advertising its network
auto-cost reference-bandwidth <Mbps>	Change the reference bandwidth used in cost calculation (must match across the OSPF domain)
ip ospf cost <value>	Manually override the calculated cost on an interface
ip ospf priority <0-255>	Set DR/BDR election priority on an interface; 0 = never become DR/BDR
default-information originate	Have this router advertise a default route into OSPF
show ip ospf neighbor	List neighbors and their state - confirm FULL (or 2WAY for DROthers)
show ip ospf interface brief	Quick view of OSPF-enabled interfaces, area, cost, and neighbor count
show ip protocols	Verify which networks/interfaces are running OSPF and timers in use
show ip route ospf	Display only OSPF-learned routes in the routing table

Configuration Walkthrough

Basic single-area OSPF on R1

```
R1(config)#router ospf 1
R1(config-router)#router-id 1.1.1.1
R1(config-router)#network 10.0.0.0 0.0.0.3 area 0
R1(config-router)#network 192.168.1.0 0.0.0.255 area 0
R1(config-router)#passive-interface GigabitEthernet0/1
```

Multi-area OSPF with route summarization at the ABR

```
R2(config)#router ospf 1
R2(config-router)#network 10.1.0.0 0.0.255.255 area 1
R2(config-router)#network 10.0.0.0 0.0.0.3 area 0
R2(config-router)#area 1 range 10.1.0.0 255.255.0.0
```

Tuning reference bandwidth for multi-gigabit links

```
R1(config)#router ospf 1
R1(config-router)#auto-cost reference-bandwidth 10000
```

Worked Scenario

Q: Two routers on the same Ethernet segment are stuck at 2-Way and never reach Full. Is this a problem?

Not necessarily. On broadcast/NBMA segments, only the DR and BDR form Full adjacencies with every other router; two DROther routers correctly stay at 2-Way with each other. If you expected Full (e.g., on a point-to-point link, or between a DROther and the DR), check that area IDs, subnet masks, Hello/Dead timers, and authentication settings match on both ends - any mismatch prevents progression past 2-Way.

Common Mistakes / Gotchas

- Forgetting the wildcard mask is inverted from a subnet mask (0.0.0.255 matches a /24, not 255.255.255.0)
- Assuming the process ID must match between routers - it's locally significant only; area IDs and timers must match, not process IDs
- Mismatched reference-bandwidth across routers, causing inconsistent cost calculations and suboptimal path selection
- Forgetting that area 0 (the backbone) must exist and all other areas must touch it directly or via a virtual link

Exam Tips

★ *Memorize the exact neighbor state sequence - it is tested directly, often as an ordering question*

★ *Cost = reference bandwidth / interface bandwidth; lower cumulative cost wins, NOT hop count*

★ *Know which LSA type is generated by which router role (ABR=Type 3, ASBR=Type 5)*